

The Potential Outcome Framework

PS200C, Causal Inference, Spring 2026

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What is causality?

What do we mean when we say X causes Y , or Y was caused by X ?

Thoughts?

Reading: Chapters 1 and 2 from "What If" (Hernan & Robins)

Flavors of causality

Effect of causes. One kind of causal question the most common type of causal question in applied areas-regards the *effect* of some potential cause, e.g. what is the effect of:

- political institutions on corruption?
- effect of a drug or other therapy on a health outcome
- an ad on purchasing behavior
- peace-keeping missions on peace?
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Other/more general/complicated concepts can be embedded in a deep enough framework (namely structural causal models)

Two languages

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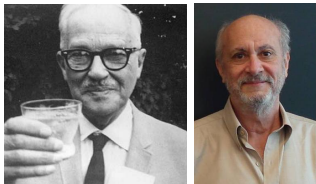


Figure: Neyman and Rubin



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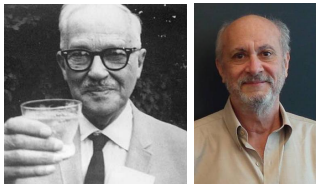


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We will use both extensively, but start with potential outcomes.

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We'll use this example throughout to make the abstract machinery concrete.

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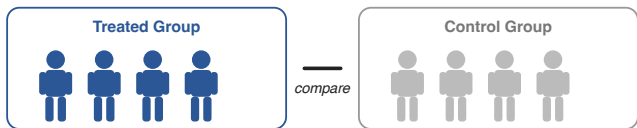
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The first is a comparison between groups of different people. The second is a comparison within each person, between two versions of their life.

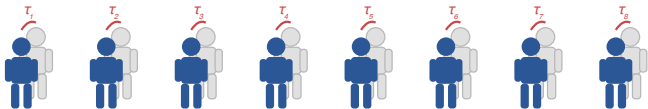
What Causal Inference Actually Asks

Group
comparison



Contrasting two groups

Causal
(counterfactual)
comparison



Contrasting two (potential) outcomes for everybody



= treated



= untreated

τ_i = individual contrast

Potential Outcomes Notation



- $Y_i(1)$: the outcome unit i *would have* under treatment
- $Y_i(0)$: the outcome unit i *would have* under control
- $\tau_i = Y_{1i} - Y_{0i}$: the causal effect for unit i
- Sample average causal effect: $\frac{1}{n} \sum_{i=1}^n \tau_i = \frac{1}{n} \sum_{i=1}^n (Y_{1i} - Y_{0i})$

The **fundamental problem of causal inference**: you only get to see *one* potential outcome per unit. We need assumptions to somehow "fill in the missing" potential outcome.

From Potential Outcomes to Observed Data

Now add the treatment indicator D_i (1 if treated, 0 if not). Key insight: *treatment doesn't change the potential outcomes—it changes which one you see:*

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- 4 τ_i

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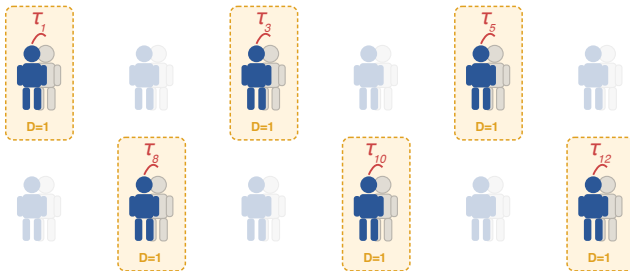
- Average treatment effect for sub-groups ($ATE(X)$):

$$ATE(X) = \mathbb{E}[Y_i(1) - Y_i(0) | X_i = x]$$

Visualizing the ATT

The ATT asks: among units who were actually treated ($D_i = 1$), what is the average of their individual causal effects?

Not a group comparison—imagine each treated unit's two potential lives:



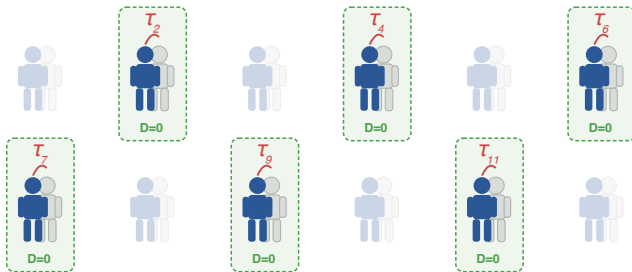
ATT = average of

$T_1, T_3, T_5, T_8, T_{10}, T_{12}$

Visualizing the ATC

The ATC asks the same question for the untreated ($D_i = 0$):

These units were *not* treated, but we still ask what *would have happened* if they had been:



ATC = average of

$T_2, T_4, T_6, T_7, T_9, T_{11}$

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You will encounter **SUTVA** (Rubin 1980) = no interference + “one version of treatment.”

- These are threats to consistency, *not additional assumptions*, don't go listing assumptions for no reason.

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Similarly, a concept may seem modifiable yet we may be unclear about what exactly it would change (e.g. the effect of obesity on health).

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- What is the direction of bias, and why?

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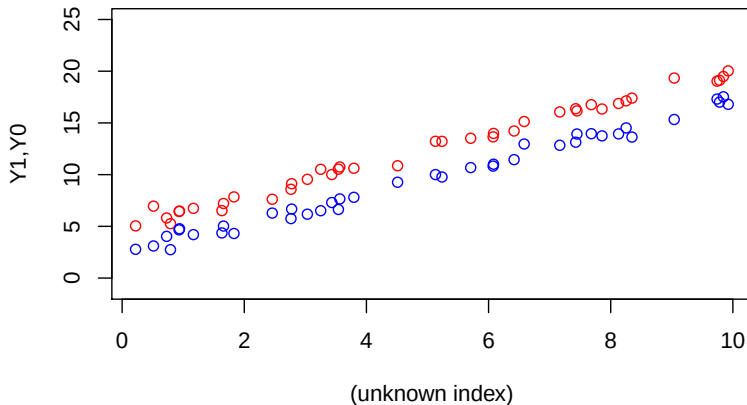
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Which of these scenarios are we usually in with observational data?

Visual Practice

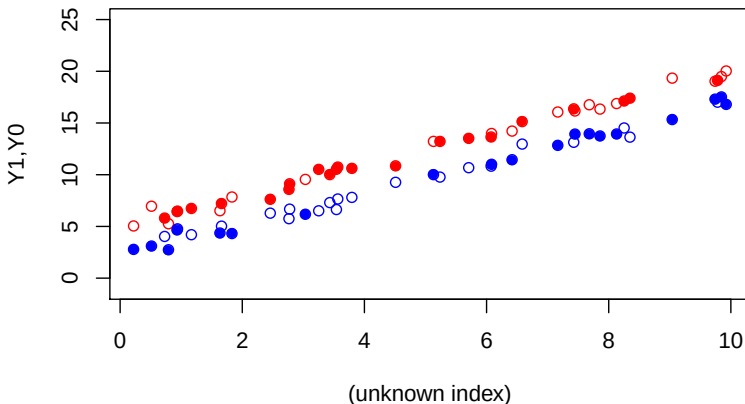
Let's look at Y_{1i} and Y_{0i} alone first.



True $\mathbb{E}[Y_{1i} - Y_{0i}] = ATE = 3$

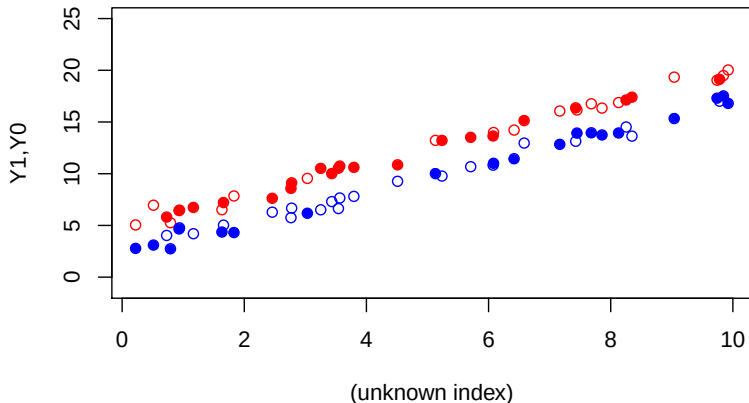
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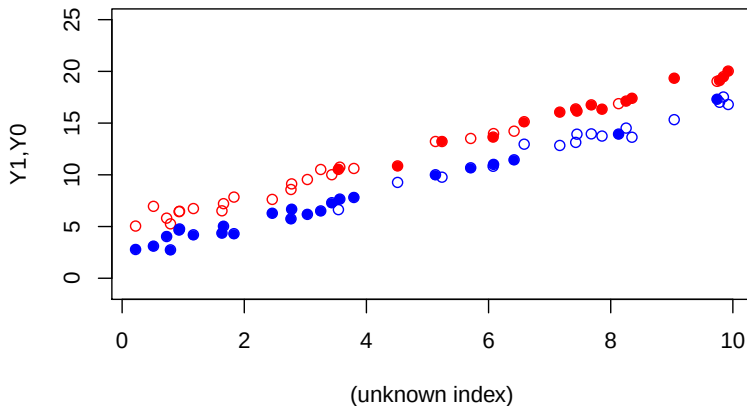


We get

$$\hat{\mathbb{E}}[Y_i | D_i = 1] - \hat{\mathbb{E}}[Y_i | D_i = 0] = \widehat{ATE} = 3.01$$

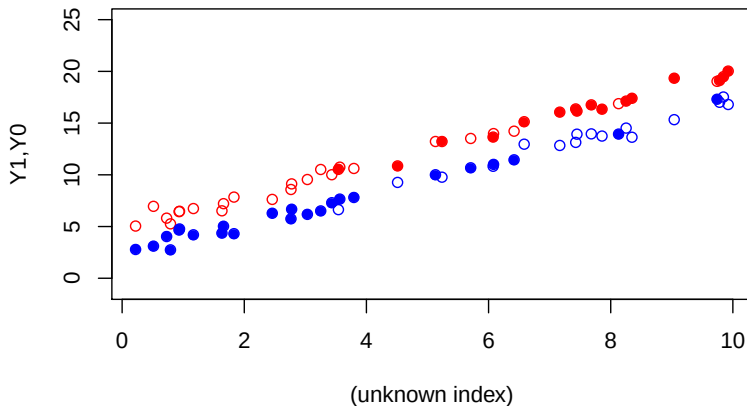
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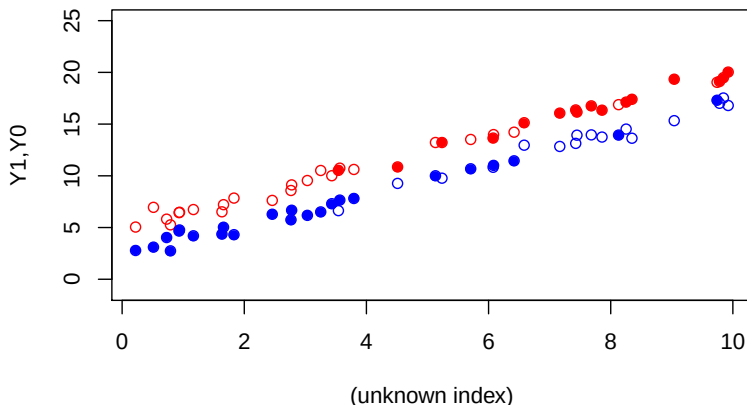
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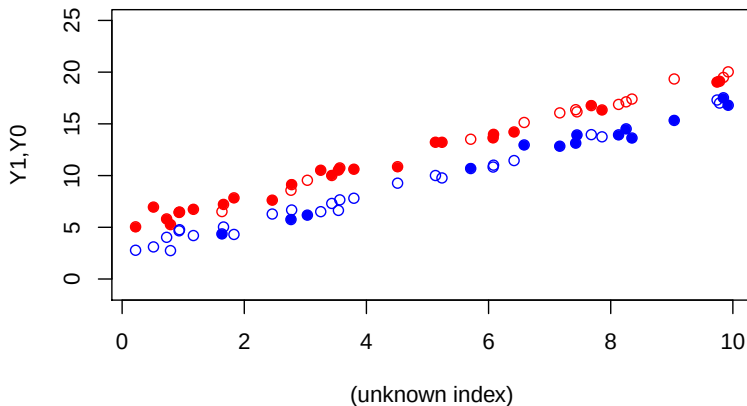


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$$\hat{\mathbb{E}}[Y_i | D_i = 1] - \hat{\mathbb{E}}[Y_i | D_i = 0] = \widehat{ATE} = 9.97$$

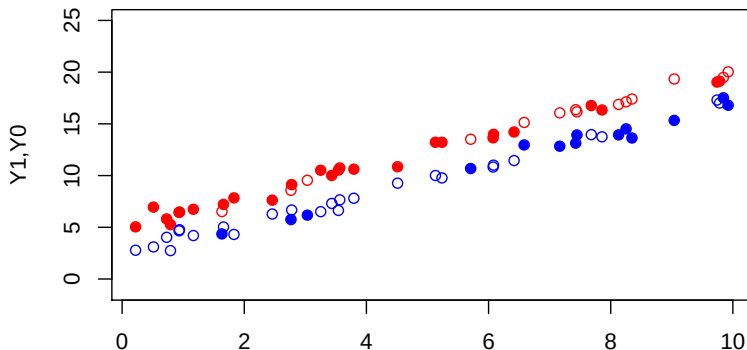
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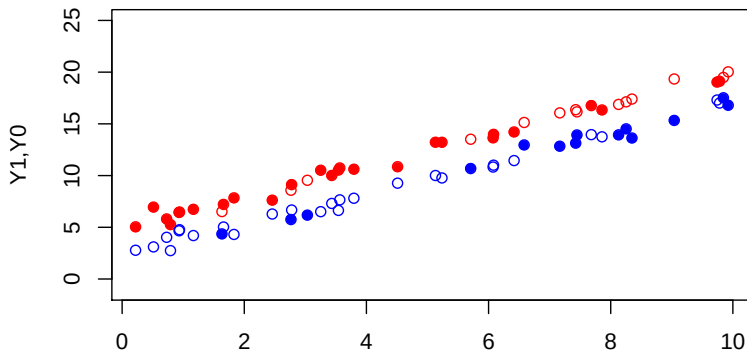


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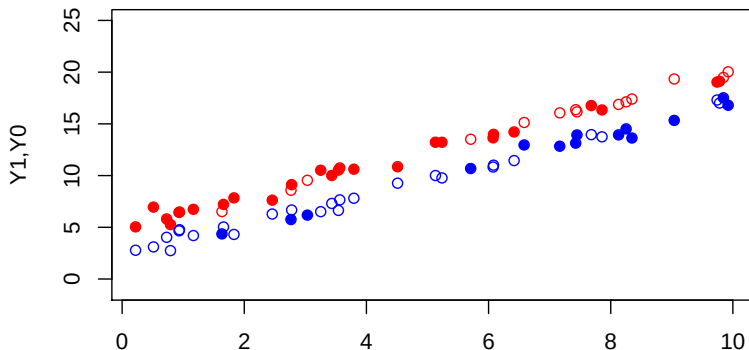
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$$\hat{\mathbb{E}}[Y_i | D_i = 1] - \hat{\mathbb{E}}[Y_i | D_i = 0] = \widehat{ATE} = -1.59$$

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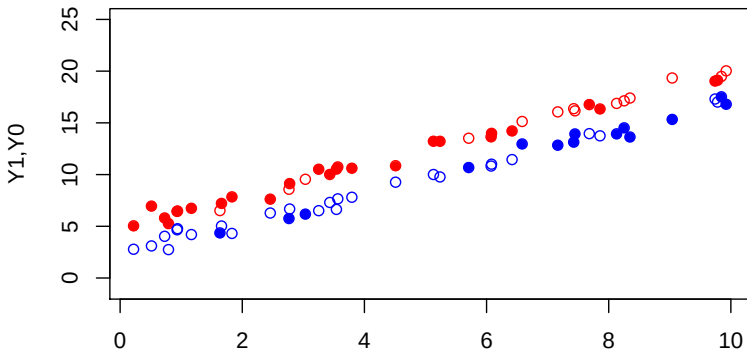
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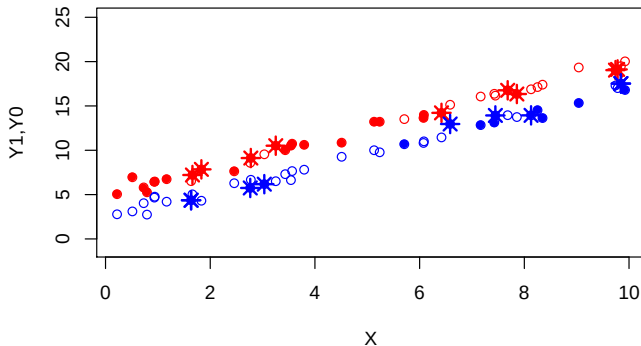
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What can we do about this? **Nothing, yet**

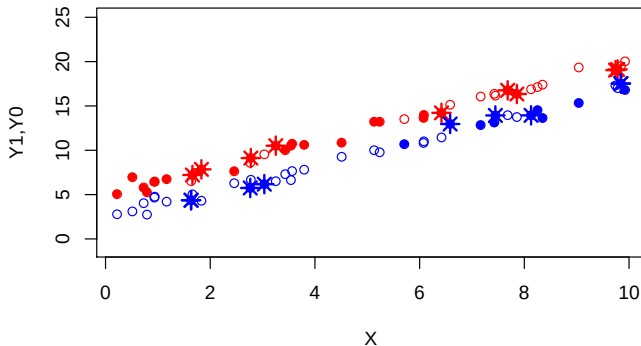
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$$[\bar{Y}_i|D=1] - [\bar{Y}_i|D=0] = \widehat{ATE} = 2.63$$

More PO practice: the “science table”

Imagine a study population with 4 units:

i	D_i	Y_{1i}	Y_{0i}	τ_i
1	1	10	4	6
2	1	1	2	-1
3	0	3	3	0
4	0	5	2	3

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2. What are the ATT and ATC?

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2	1	1	2	-1
3	0	3	3	0
4	0	5	2	3

1. What is the ATE? $\mathbb{E}[Y_{1i} - Y_{0i}] = 1/4 \times (6 - 1 + 0 + 3) = 2$

(Note: average effect is positive, but not all τ_i are)

2. What are the ATT and ATC?

$$\mathbb{E}[Y_{1i} - Y_{0i} | D_i = 1] = .5(6 - 1) = 2.5$$

More PO practice: the “science table”

Imagine a study population with 4 units:

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What are the terms in **blue** & **green**?

Selection bias: examples

Recall: $\text{DIM} = \text{ATT} + \text{selection bias}$, where
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Our running example: D = College degree, Y = Liberal attitudes

- People who get degrees may already be more liberal (parental education, urban upbringing). So $\text{bias} \neq 0$.

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Those represent important assignment mechanisms with hallowed names in varied fields:

- random assignment, ignorability, or exchangeability
- selection on observables, conditional ignorability
- bad news

What's next

From [here](#): random assignment, then
random-conditional-on-observables.